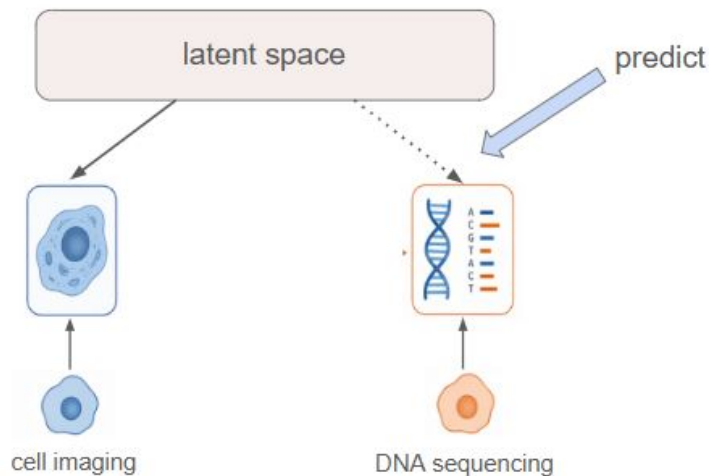


Causal Representation Learning for Cross-Modal Transfer

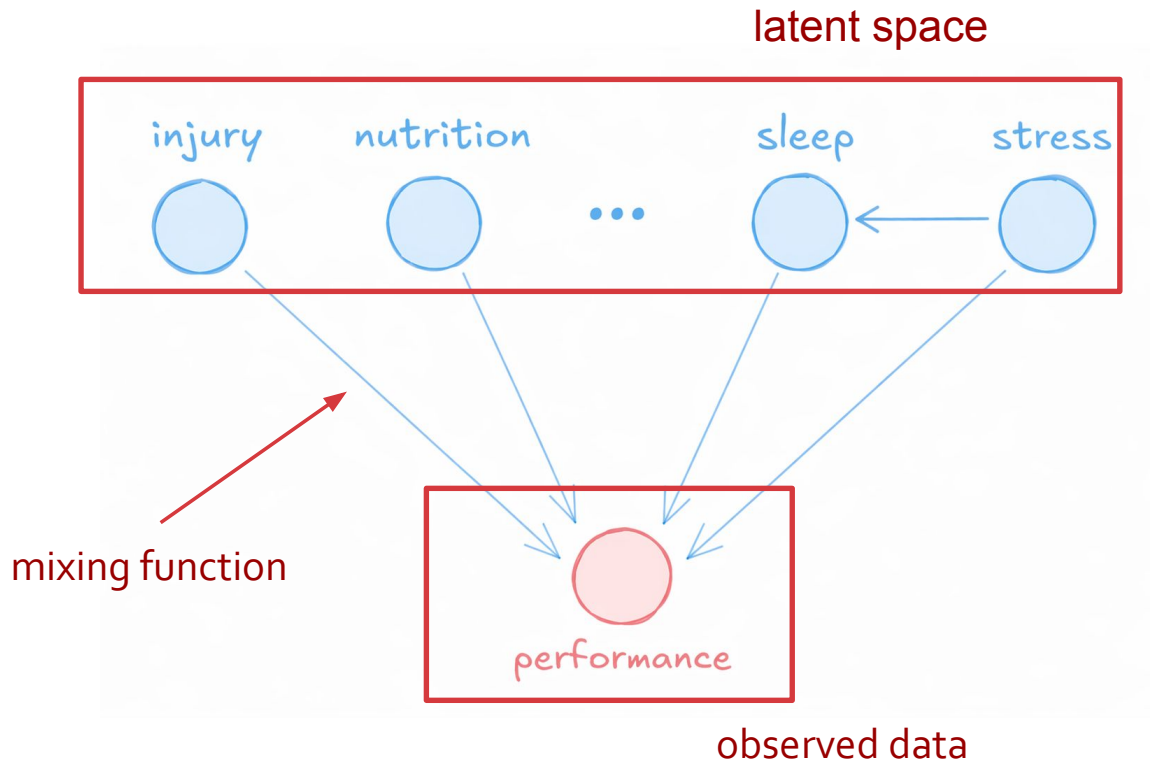
Nikolija Bojkovic, Abhinav Kumar, Prof. Caroline Uhler



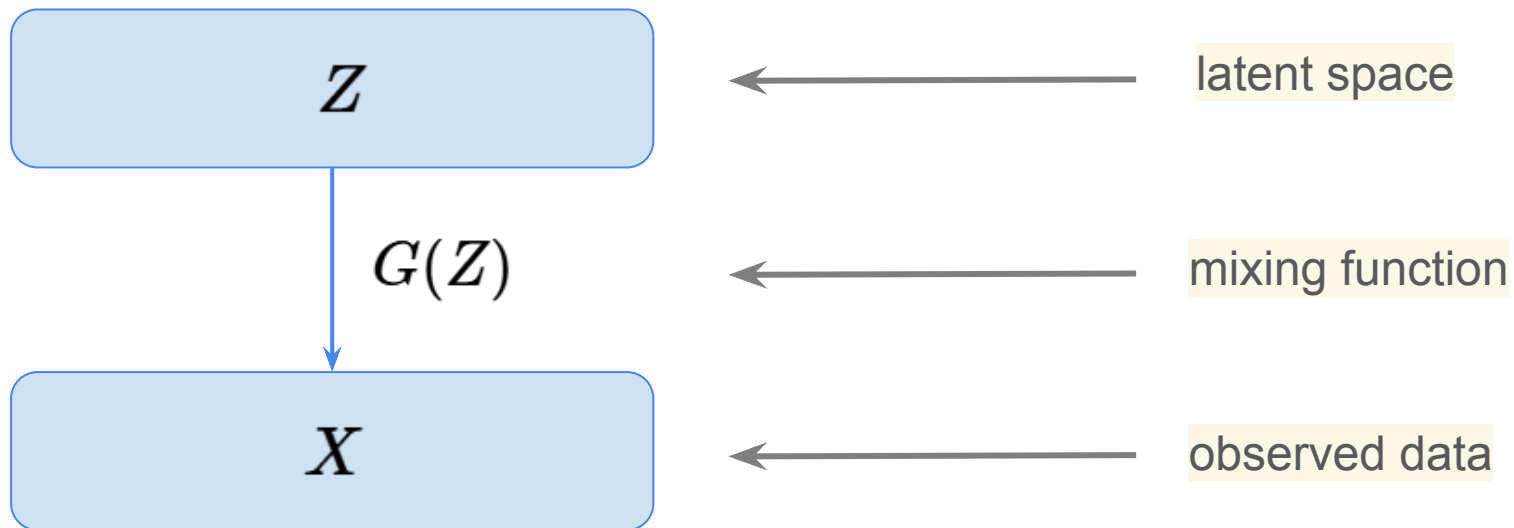
Simple Example

Simple Analogy: Athlete Performance

Can we recover the latent space only from observed data?



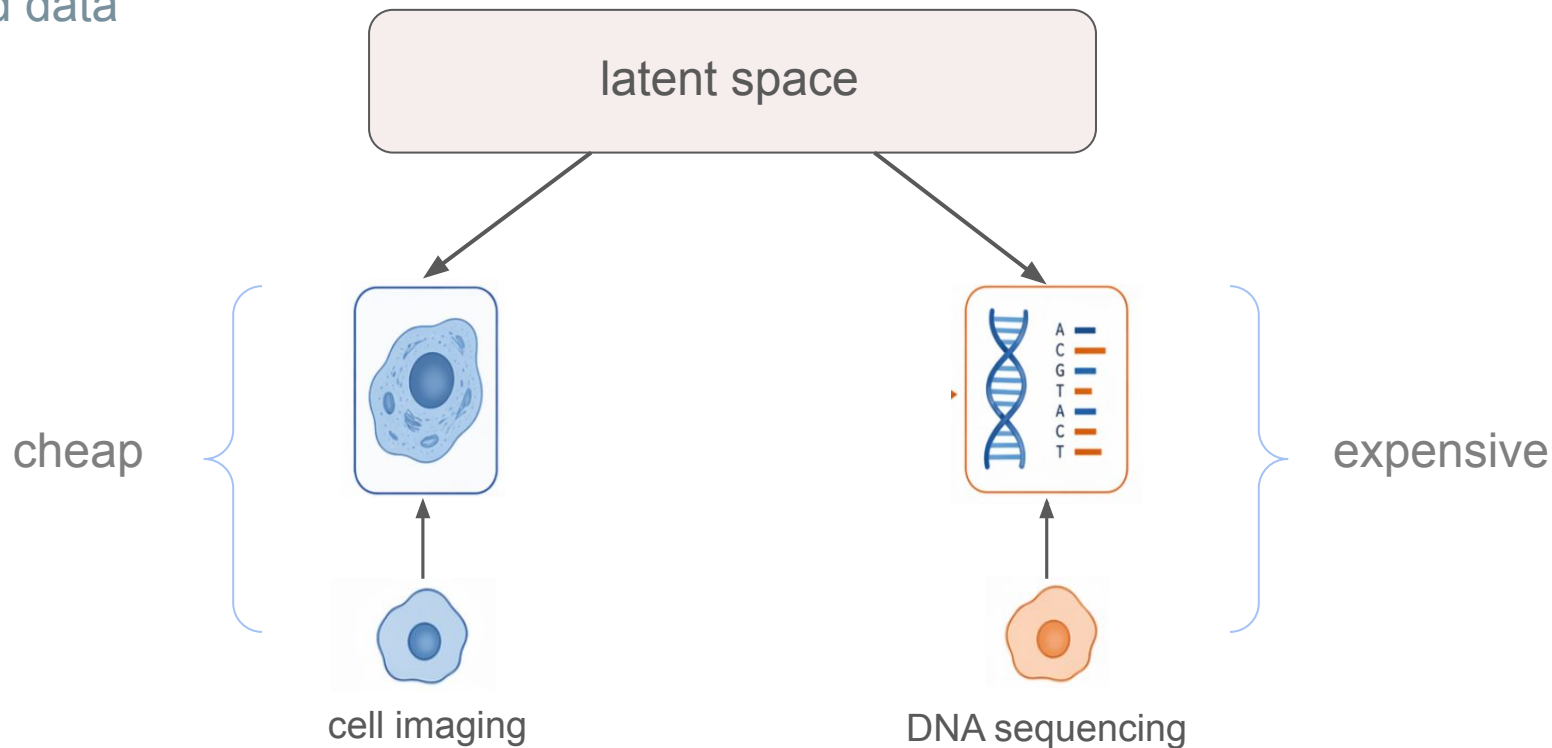
CRL Framework



Motivating Example

Can we use a cheap modality to predict an expensive one?

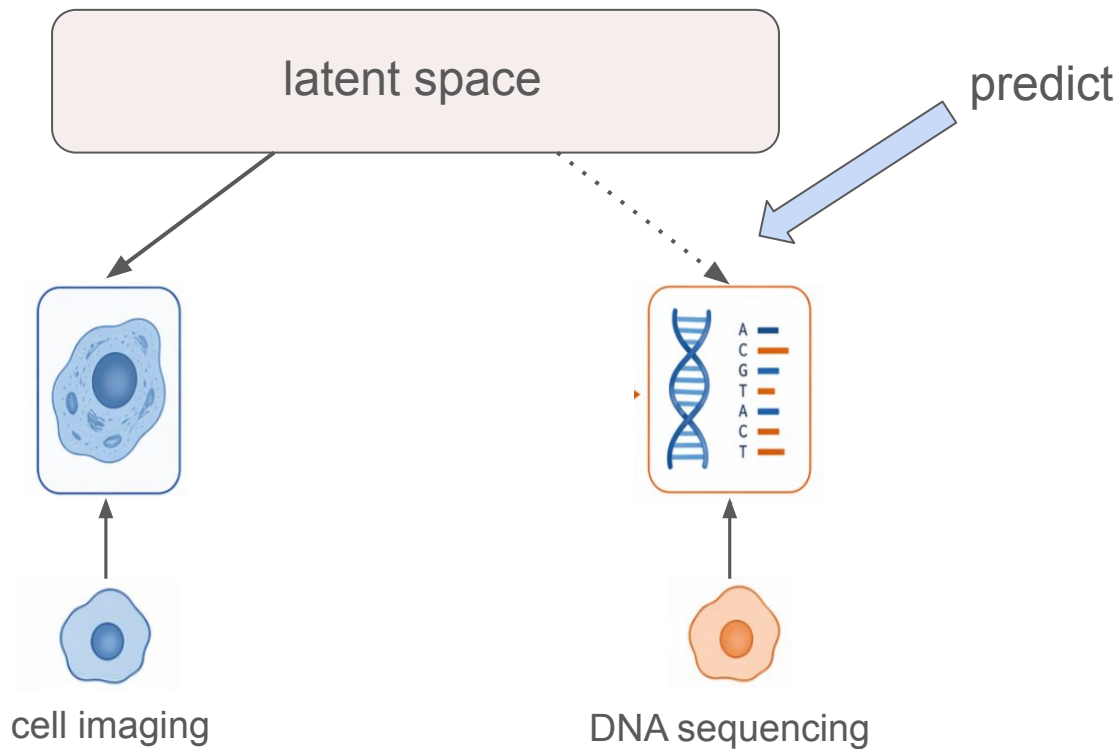
Paired data



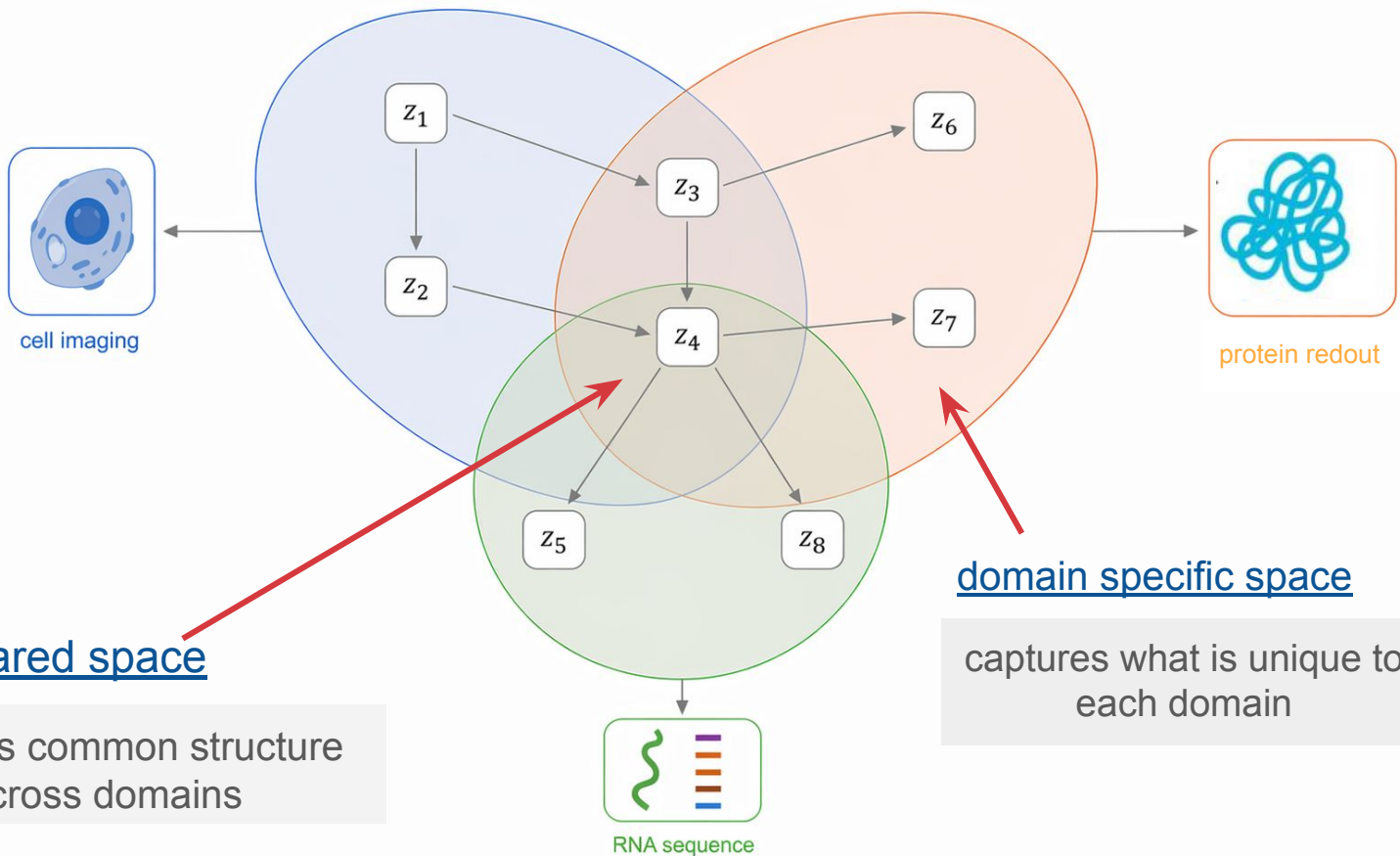
Motivating Example

Replace expensive measurements with predictions
based on cheaper ones

Paired data

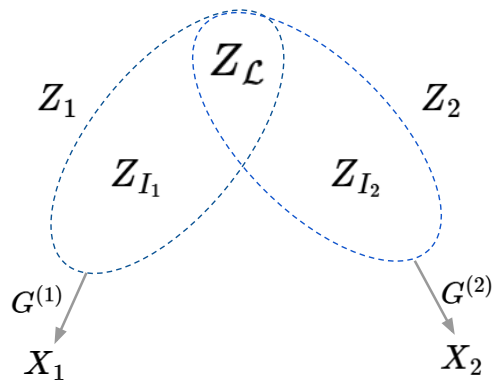


Problem Setup



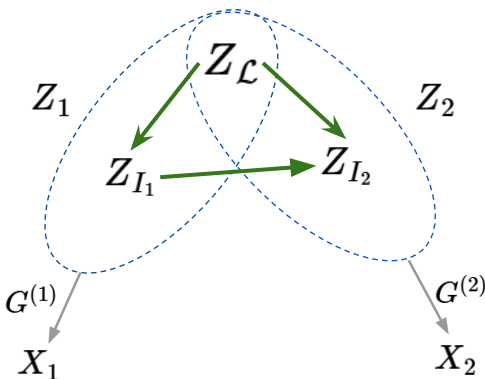
Comparison with Previous Work

Sturma et al. (2023)



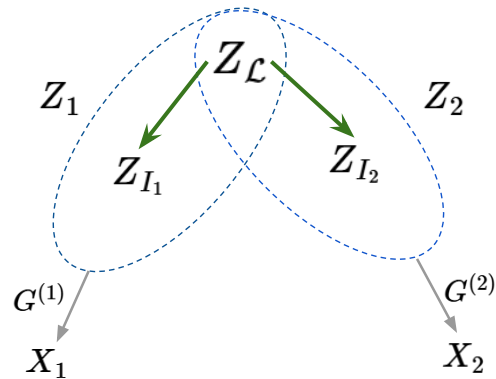
- Multi-Domain
- Latent structural assumptions
- Linear mixing
- Unpaired data

Yao et al. (2024)



- Multi-Domain
- No latent structural assumptions
- General mixing
- Paired data

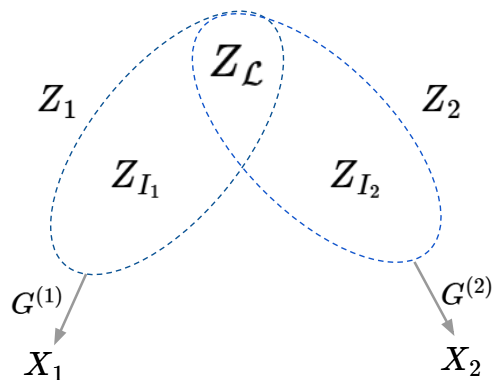
Our work



- Multi-Domain
- Partial latent structural assumptions
- Polynomial mixing
- Paired data

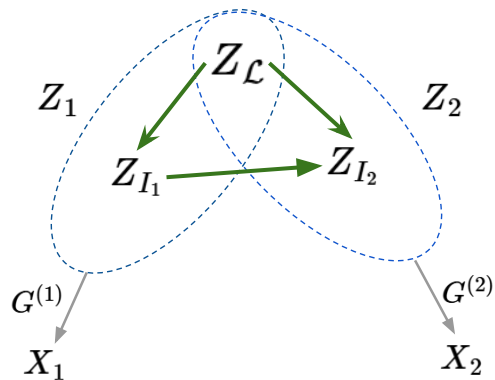
Comparison with Previous Work

Sturma et al. (2023)



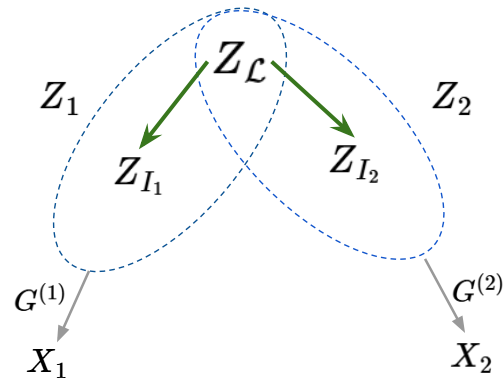
- full recovery under strong structural assumption

Yao et al. (2024)



- shared space up to bijection
- domain-specific given independence from the shared

Our work



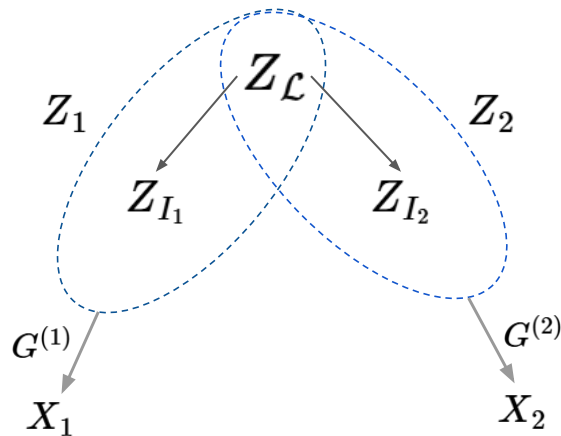
- shared space up to affine transformation
- domain-specific space in distribution

Result I: Can we identify the dimension of the shared space?

Identification of Latent Dimensions

Assumption:

G^e is a full-row-rank polynomial for each $e \in \{1, 2\}$



Lemma I (informal): We can recover:

$$|Z_{\mathcal{L}}| \quad |Z_1| \quad |Z_2|$$

Previous Work

- Zhang et al. (2023) • Single-domain setting with unpaired interventional data

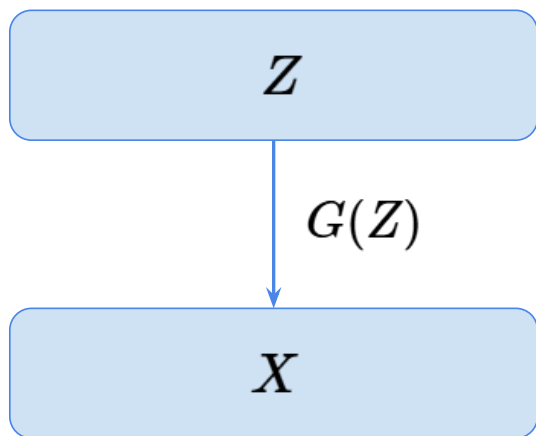
$G(Z)$ full-row-rank polynomial

identifiable up to affine transformation!

$$\hat{Z} = Z\Lambda + b$$

+

$|Z|$ is identifiable



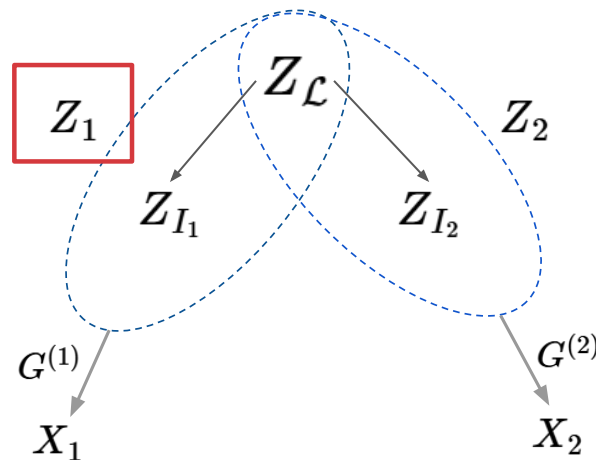
Proof Sketch

Zhang et al. (2023)

$$\hat{Z} = Z\Lambda + b$$



$$\underbrace{\begin{pmatrix} \hat{Z}_{\mathcal{L},1} \\ \hat{Z}_{I_1} \end{pmatrix}}_{\hat{Z}_1} = \begin{pmatrix} A_1 & B_1 \\ C_1 & D_1 \end{pmatrix} \underbrace{\begin{pmatrix} Z_{\mathcal{L}} \\ Z_{I_1} \end{pmatrix}}_{Z_1} + \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}$$



Proof Sketch

Zhang et al. (2023)

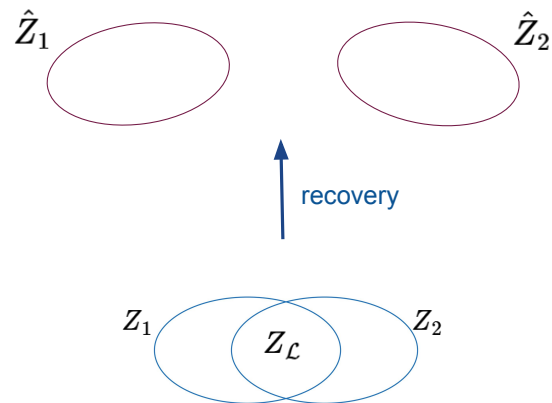
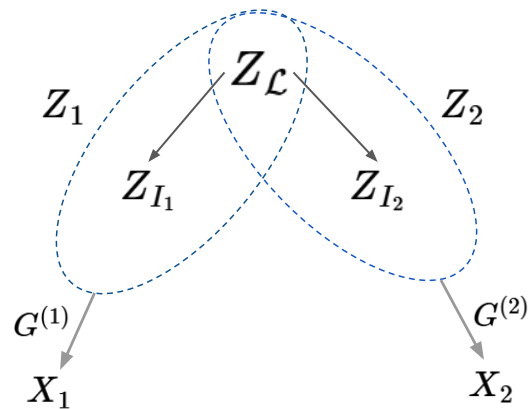
$$\hat{Z} = Z\Lambda + b$$



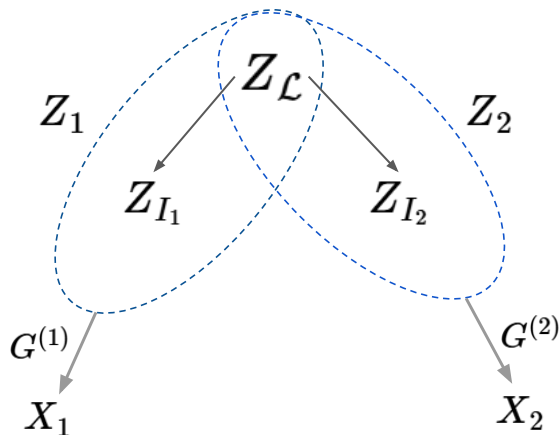
$$\begin{pmatrix} \hat{Z}_{\mathcal{L},1} \\ \hat{Z}_{I_1} \end{pmatrix} = \begin{pmatrix} A_1 & B_1 \\ C_1 & D_1 \end{pmatrix} \begin{pmatrix} Z_{\mathcal{L}} \\ Z_{I_1} \end{pmatrix} + \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}$$

$$\begin{pmatrix} \hat{Z}_{\mathcal{L},2} \\ \hat{Z}_{I_2} \end{pmatrix} = \begin{pmatrix} E_1 & F_1 \\ G_1 & H_1 \end{pmatrix} \begin{pmatrix} Z_{\mathcal{L}} \\ Z_{I_2} \end{pmatrix} + \begin{pmatrix} r_1 \\ r_2 \end{pmatrix}$$

$$\hat{Z} = Z\Lambda + c$$



Proof Sketch



shared dimension:

$$|Z_{\mathcal{L}}| = |Z_1| + |Z_2| - |Z|$$

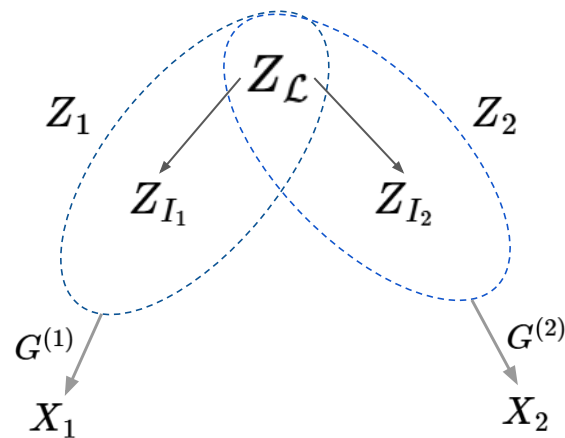
modality specific
dimensions:

$$|Z_{I_1}| = |Z_1| - |Z_{\mathcal{L}}|$$

$$|Z_{I_2}| = |Z_2| - |Z_{\mathcal{L}}|$$

Result II: Can the shared latent variable depend on domain-specific recovered coordinates?

$$\underbrace{\begin{pmatrix} \hat{Z}_{\mathcal{L},1} \\ \hat{Z}_{I_1} \end{pmatrix}}_{\hat{Z}_1} = \begin{pmatrix} A_1 & B_1 \\ C_1 & D_1 \end{pmatrix} \begin{pmatrix} Z_{\mathcal{L}} \\ Z_{I_1} \end{pmatrix} + \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}$$



Theorem I (informal): The shared latent variable doesn't depend on domain-specific recovered coordinates

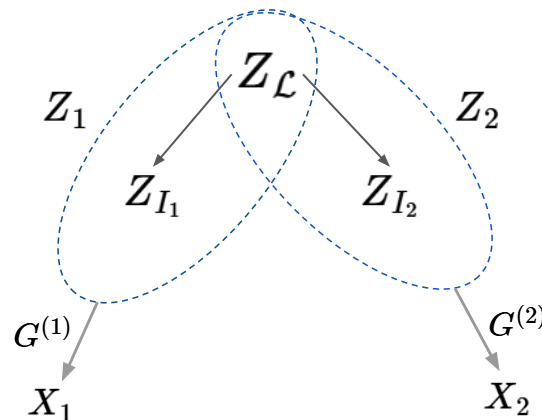
$$\hat{Z}_{\mathcal{L},1} = A_1 Z_{\mathcal{L}} + b_1$$

Proof Sketch

we proved: $B_1 = 0$

$$\mathbf{P} \begin{pmatrix} \hat{Z}_{\mathcal{L},1} \\ \hat{Z}_{I_1} \end{pmatrix} = \underbrace{\begin{pmatrix} A_1 & B_1 \\ C_1 & D_1 \end{pmatrix}}_{\text{full rank}} \begin{pmatrix} Z_{\mathcal{L}} \\ Z_{I_1} \end{pmatrix} + \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}$$

$$\hat{Z}_{\mathcal{L},1} = A_1 Z_{\mathcal{L}} + b_1$$



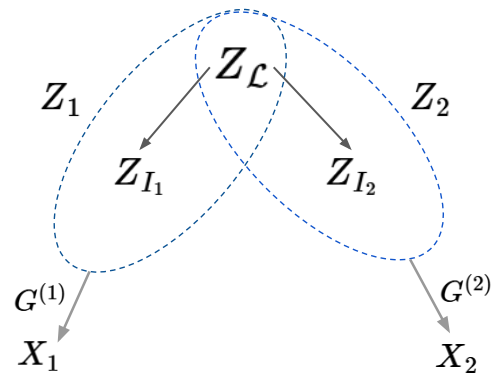
Wait! Does this mean we have recovered the shared latent space??

not really! :(

Result III: Can we actually recover the shared space itself?

Identification of Shared Space

Assumption: $Z_{I_1} \perp Z_{I_2} \mid Z_{\mathcal{L}}$



Theorem I (informal): We can identify which coordinates correspond to:

$$Z_{\mathcal{L}}$$

and the private and shared coordinates of:

$$\hat{Z}_1$$

$$\hat{Z}_2$$

Identification of Shared Space

$$\Sigma_{1|2} := \mathbb{E} \left[\text{Cov}(\hat{Z}^{(1)} \mid \hat{Z}^{(2)}) \right]$$

$$\hat{\Sigma}_{1|2} = P^T \begin{pmatrix} \boxed{0_{p \times p}} & 0 \\ 0 & \boxed{D_1 \text{Cov}(Z_{I_1} \mid Z_{\mathcal{L}}) D_1^\top} \end{pmatrix} P$$

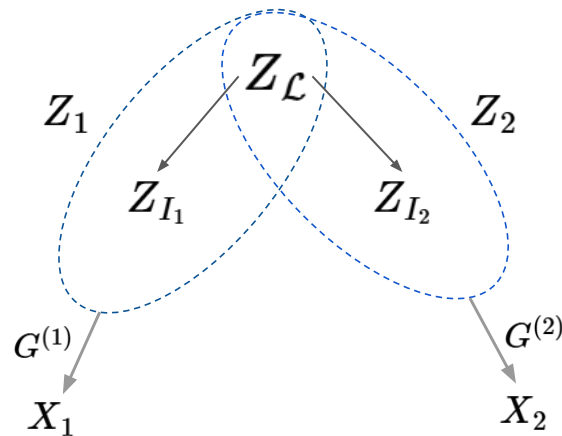
Diagram illustrating the identification of the shared space. The equation shows the estimated covariance matrix $\hat{\Sigma}_{1|2}$ as a function of the permutation matrix P (labeled "permutation matrix" with an arrow) and the observed data $Z_{\mathcal{L}}$ (indicated by an arrow pointing to the top-left block $0_{p \times p}$). The bottom-right block is $D_1 \text{Cov}(Z_{I_1} \mid Z_{\mathcal{L}}) D_1^\top$, where D_1 is the shared space matrix (indicated by an arrow pointing to the block) and $\hat{Z}_1$ is the latent variable (indicated by an arrow pointing to the block).

Identification of Shared Space

$$\Sigma_{1|2} := \mathbb{E} \left[\text{Cov}(\hat{Z}^{(1)} \mid \hat{Z}^{(2)}) \right]$$

$$\hat{\Sigma}_{1|2} = P^T \begin{pmatrix} \boxed{0_{p \times p}} & Z_{\mathcal{L}} & 0 \\ 0 & \boxed{D_1 \text{Cov}(Z_{I_1} \mid Z_{\mathcal{L}}) D_1^T} \end{pmatrix} P$$

$Z_{\mathcal{L}}$ (pointing to the top-left block)
permutation matrix (pointing to P)
 $\hat{Z}_1$ (pointing to the bottom-right block)



$$\boxed{Z_{\mathcal{L}} = E\hat{Z}_{\mathcal{L},2} + c_2} \leftarrow \text{fully determined by } \hat{Z}_2 = \begin{pmatrix} \hat{Z}_{\mathcal{L},2} \\ \hat{Z}_{I_2} \end{pmatrix}$$

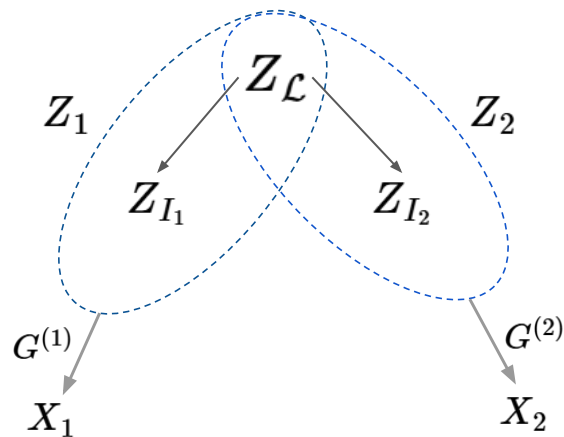
Result IV: How well can we recover the domain-specific latent variables having recovered latent shared representation?

Identification of Domain-Specific Space

Assumption I: $Z_{I_1} \perp Z_{I_2} \mid Z_{\mathcal{L}}$

Assumption II:

G^e is a full-row-rank polynomial for each $e \in \{1, 2\}$



Theorem III (informal): The distribution of private variables is identifiable only up to an invertible **affine transformation**

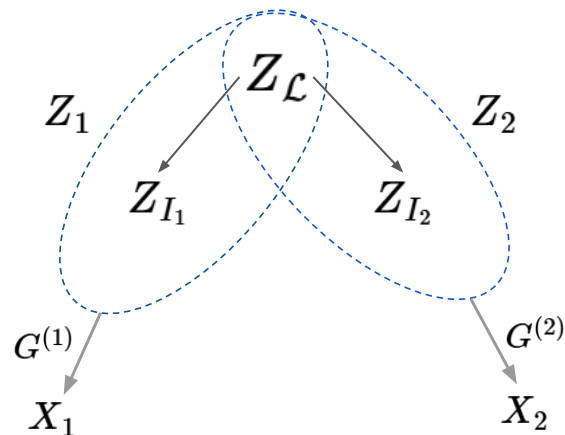
Proof Sketch

- Fix the shared part: $Z_{\mathcal{L}} = z$



$$\hat{Z}_{I_1} = C_1 A_1^{-1}(\hat{Z}_{\mathcal{L}} - b_1) + D_1 Z_{I_1} + b_2$$

$$Z_{I_1} = D_1^{-1}(\hat{Z}_{I_1} - c(z))$$



Any other valid solution differs only by another invertible affine map



the private distribution is identifiable only up to affine transformation

Gaussian Case

Objective:

$$\min_{(\mathbb{P}_Z, \hat{G}) \in \mathcal{F}_{\hat{p}}} \hat{p} \quad \text{subject to} \quad \mathbb{P}_{\hat{G}(Z)} = \mathbb{P}_X$$

Algorithm 1 Greedy Latent Dimension Recovery (Gaussian)

```

1: for  $\hat{p} = 1, 2, 3, \dots, p_{\max}$  do
2:   Initialize  $\mu \leftarrow 0$ ,  $L \leftarrow I_{\hat{p}}$ ,  $G \leftarrow$  small random
3:   for  $t = 1, \dots, T$  do
4:     Sample  $\epsilon \in \mathbb{R}^{n \times \hat{p}}$ , rows i.i.d.  $\mathcal{N}(0, I_{\hat{p}})$ 
5:      $\hat{Z} \leftarrow \mu + \epsilon \text{tril}(L)^\top$ 
6:      $\hat{X} \leftarrow \varphi(\hat{Z})G^\top$ , then standardize  $\hat{X}$ 
7:      $\mathcal{L} \leftarrow \text{MMD}^2(X, \hat{X})$ 
8:     Update  $\mu$ ,  $L$ ,  $G$  via Adam
9:   end for
10:  if  $\mathcal{L} < \tau$  and  $\text{rank}(G) = n_\varphi$  then
11:    return  $\hat{p}$ 
12:  end if
13: end for

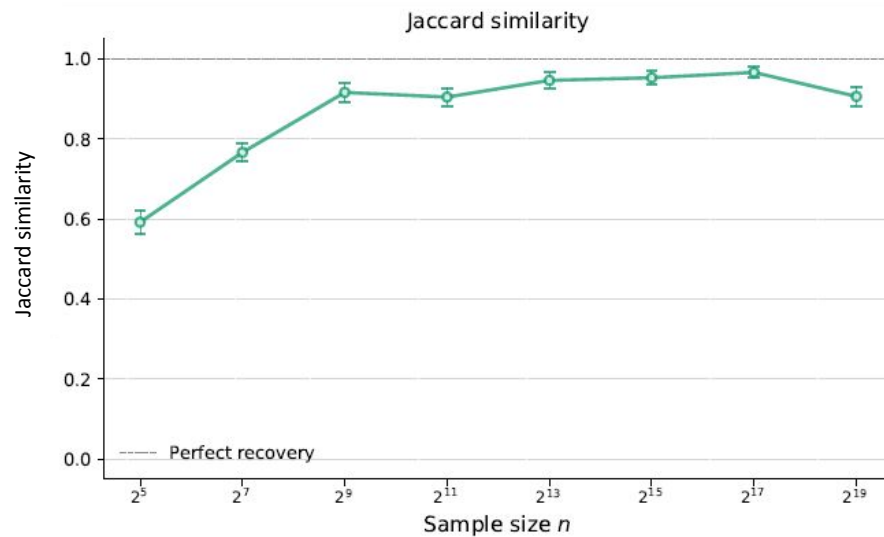
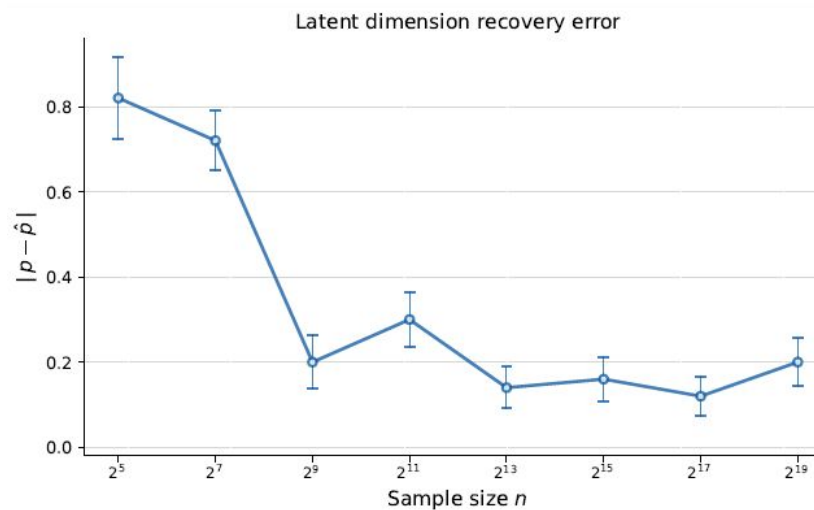
```

try candidate latent dimensions

sample latent variables using
Gaussian model

generate reconstructed
observations

Gaussian Case



General Case

Algorithm 3 Greedy Latent Dimension Recovery (General)

```

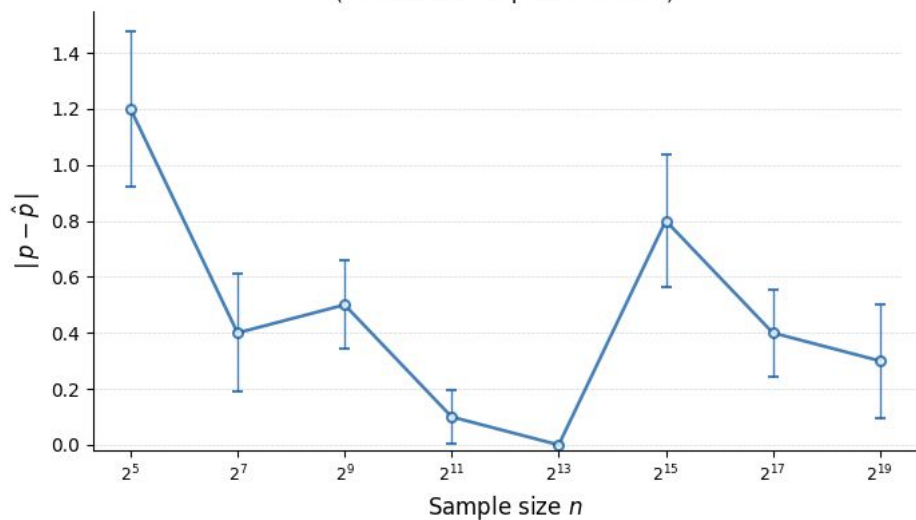
1: for  $\hat{p} = 1, 2, 3, \dots, p_{\max}$  do
2:   Initialize normalizing flow  $f_{\hat{p}}$  (RealNVP, 6 layers, hidden dim 64),  $G \leftarrow$  small random
3:   for  $t = 1, \dots, T$  do
4:     Sample  $\epsilon \in \mathbb{R}^{n \times \hat{p}}$ , rows i.i.d.  $\mathcal{N}(0, I_{\hat{p}})$ 
5:      $\hat{Z} \leftarrow f_{\hat{p}}(\epsilon)$ 
6:      $\hat{X} \leftarrow \varphi(\hat{Z})G^{\top}$ , then standardize  $\hat{X}$ 
7:      $\mathcal{L} \leftarrow \overline{\text{MMD}}^2(\hat{X}, X)$ 
8:     Update  $f_{\hat{p}}, G$  via Adam with cosine annealing
9:   end for
10:  if  $\mathcal{L} < \tau$  and  $\text{rank}(G) = n_{\varphi}$  then
11:    return  $\hat{p}, f_{\hat{p}}$ 
12:  end if
13: end for

```

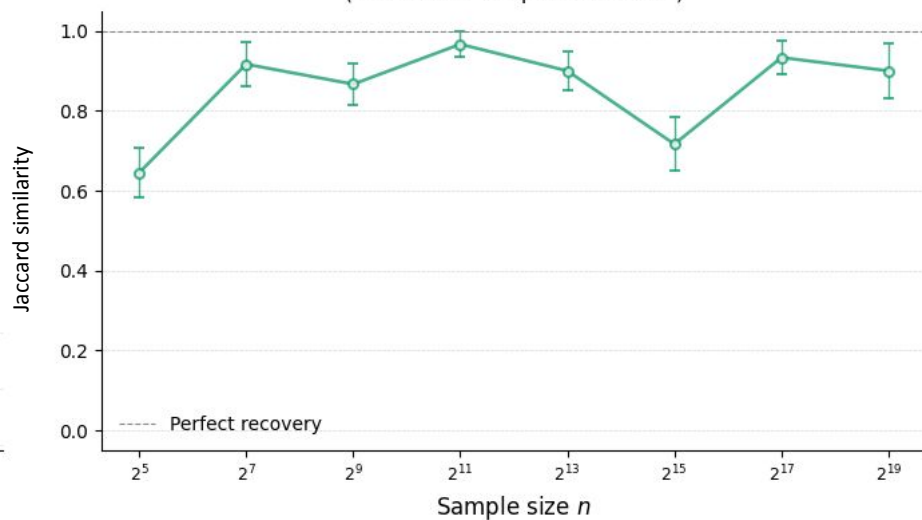
General Case

In progress...

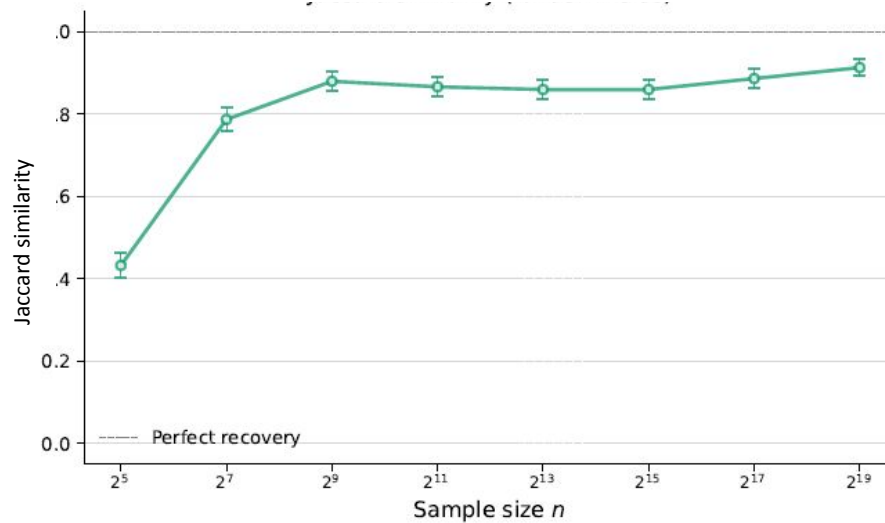
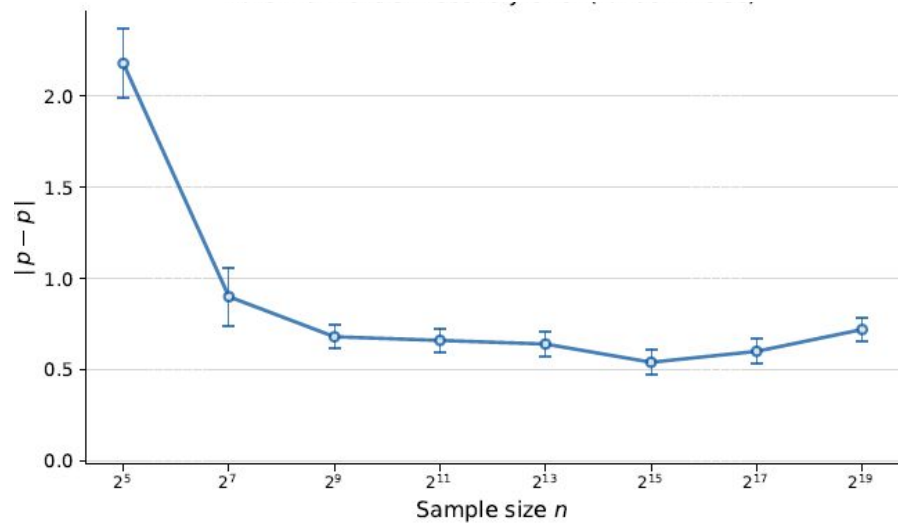
Latent dimension recovery error
(Theorem 3 — exponential basis)



Jaccard similarity
(Theorem 3 — exponential basis)



Gaussian Case



Future work

- General Mixing Function
- Experiments on real-world data
- Reducing the structure assumption

Thank you for your attention!

Simulations

- DAG, SCM
- A flow function starts from simple Gaussian noise and transforms it through an invertible nonlinear map into a much more general latent distribution.